# Norman Rasmussen

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**BS, Applied & Computational Mathematics Emphasis (ACME)** **Apr 2027**
*Brigham Young University* *Provo, UT*

- Concentration: Operations Research
- GPA 3.95
- Academic Scholarship, Dean's List
- Relevant Coursework: Stochastic Processes, Neural Nets, Machine Learning, Reinforcement Learning, Game Theory, Dynamic Programming, Probability Theory, Linear/Integer Programming, Computational Programming, Linear and Nonlinear Analysis, Statistics, Optimization, Data Structures

## SKILLS

Proficient in - Python (NumPy, Matplotlib, SciPy, pandas, PyTorch, scikit-learn, OR-Tools), Git, Linux, LaTeX, SQL, Excel, SQL Server Management
Experience with - C++, reStructuredText, Bash, Docker, Gurobi
Can speak, read, and write - French

## EXPERIENCE

**Research Assistant - Operations Research** **April 2025-Present**
*BYU IDeA Labs* *Provo, UT*

- Construct a supply chain model that can model any supply chain network for a series of demands, shows the existence of the Bullwhip Effect, and quantifies the impact of demand information sharing
- Create a CP-SAT model that optimizes the ordering for any set of flights at a given airport to solve for flight ordering in under 0.2 seconds
- Gather data through 100,000+ simulations of flight delays to formulate an upper bound on the maximum difference between a priority ordering and optimal ordering

**Software Engineering Intern** **May 2026-August 2026**
*UPS Mail Innovations Marketing* *Alpharetta, GA*

- Developed, deployed, and maintained two end-to-end Python/Streamlit applications, owning the full software lifecycle from creation and development through testing, deployment, and production use.
- Built a mission-critical pricing platform used to price 100+ customers representing over $800M in opportunities, with more than $400M in business won
- Created a network analytics platform that provide first time visibility into trends, at-risk accounts, onboarding, and performance across 1,000+ customer accounts

**Data Science Coursework Creator** **April 2025-Sep 2025**
*BYU Math Department* *Provo, UT*

- Designed hands-on coding labs to teach skills: pandas, PCA, PageRank, KNN Classifier
- Made additions to a preexisting website in reST that will be used by 30+ students per year

**Research Assistant - Parameter Estimation** **Sep 2024-April 2025**
*BYU Math Department* *Provo, UT*

- Developed a novel repeated-subset estimation technique, greatly improving accuracy on common ODEs
- Created a foundation for using the framework as an equation discovery tool with JAX acceleration

## PERSONAL PROJECTS

**AeroYield - Airline Revenue Optimizer** **Python | JavaScript | SciPy | scikit-learn**

- Built an interactive airline revenue optimizer using DOT data across 100 high-volume route
- Designed dashboard to provide insights into crucial KPIs to motivate theoretical business decisions
- Compared 8 seat-control policies using DP, LP, Bayesian, robust, and neural methods

**Tic-Tac-Nope - Imperfect-Information Game Solver** **Python | JavaScript | Gurobi | SciPy**

- Built an interactive website for imperfect-information tic-tac-toe with Optimal Nash Strategies
- Computed mixed over 100+ Nash equilibria using sequence-form LPs with ~2.5M player-action sequences
- Optimized sparse LP solves with Gurobi, SciPy/HiGHS, symmetry reductions, and certificates

**Shipping OR Solver** **OR-Tools | NumPy**

- Used a CP-SAT model to build an optimal 2D truck packer that accounts for 4 different items
- Used packing model data to perform PCA that shows numerical significance of each item
- Built a Vehicle Routing Solver using a CP-SAT model that accounts for repeated routes and split orders